

Biosensing-Driven Cognitive Augmentation: Real-Time Mind Wandering Detection and Robot Intervention

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Abstract

Increased information overload and cognitive fatigue have made it increasingly difficult for individuals to sustain attention and regulate emotions in everyday life. While mindfulness practice offers a proven approach to restoring attentional focus and emotional balance, its effectiveness can be undermined by frequent mind wandering. In this paper, we thus introduce a real-time robotic mindfulness intervention system that integrates a multi-biometric mind-wandering detector with the social robot Pepper. The system continuously monitors user state by analyzing biometric indicators and provides timely, context-aware verbal and non-verbal interventions generated via a GPT-based model. Our system estimates the mind-wandering (MW) level in real-time, and if the MW level exceeds a threshold, Pepper generates personalized feedback using off-the-shelf ChatGPT. In our experimental study with 45 participants, the robot-assisted group achieved higher Mindful Attention Awareness Scale (MAAS) scores and lower MW levels than both the no-intervention and audio-only groups. MAAS and MW showed a strong negative correlation ($r = -0.78, p < .001$). These results demonstrate that the robot's embodied presence and multimodal feedback effectively support attention recovery and enhance mindfulness engagement.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computer systems organization** → *Robotics*; • **Computing methodologies** → *Machine learning*; Natural language generation; • **Applied computing** → Health informatics.

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Keywords

Mind-Wandering Detection, Multimodal Biometrics, Social Robot Intervention, Mindfulness, LLM-Based Adaptive Feedback

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1 Introduction

In modern society, individuals often experience psychological stress and struggle to maintain concentration due to information overload and complex work environments [14]. Mindfulness has been recognized as an effective strategy to alleviate these issues, and it has been proven to contribute to stress reduction, improved concentration, and emotional regulation through conscious attention to the present moment [7, 13]. Numerous reports indicate that mindfulness-based programs are effective in enhancing cognitive and emotional performance in educational and occupational contexts [15, 16]. However, users with limited meditation experience may struggle to recognize mind wandering (MW) in real-time, as their attention tends to drift involuntarily.

Recent studies have suggested the possibility of reliably detecting MW levels in natural environments using multimodal biosensing and providing immediate feedback based on this [4, 6, 11]. Furthermore, large language models (LLMs) have demonstrated the potential to support mental health interventions through context-adaptive language generation [9]. Additionally, there has been an increase in research combining LLMs with robot agents to facilitate real-time healthcare interventions [2]. Nevertheless, existing studies have limitations in that (i) detection and intervention are limited to a single channel, or (ii) they fail to comprehensively address bio-signal-based cognitive state detection and embodied feedback.

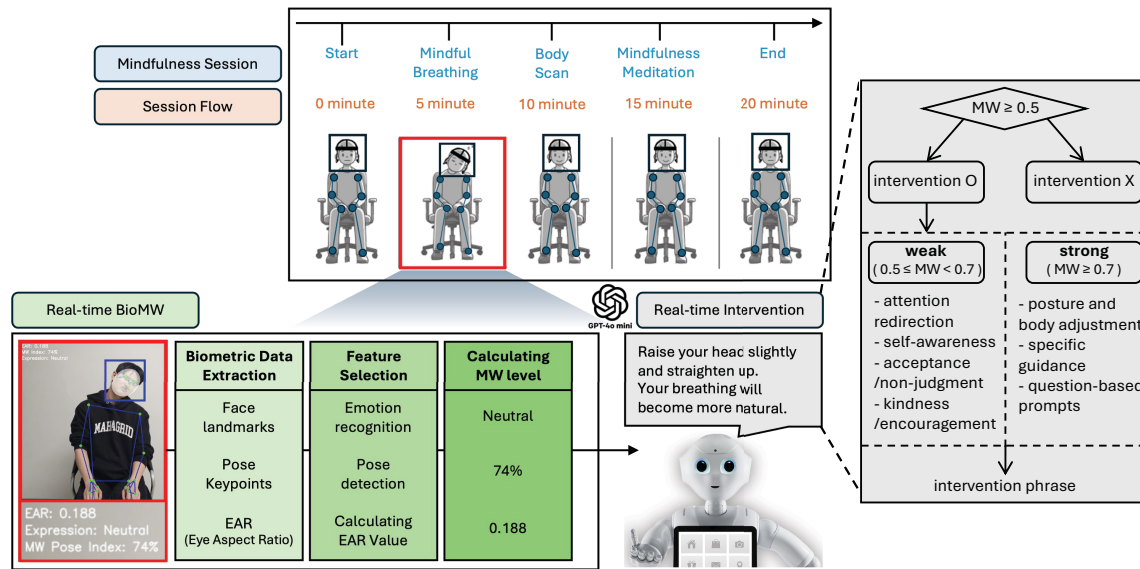


Figure 1: Overview of the real-time intervention system using Pepper robot during a mindfulness session.

This study aims to address this gap by proposing a system, illustrated in Figure 1, that integrates and analyzes non-invasive biosignals, such as facial expressions, posture, and eye aspect ratio (EAR), in real-time to detect MW. The system provides customized feedback generated by a GPT-based language model via a social robot when the threshold is exceeded. The proposed system processes camera-based biosignal streams using a lightweight multimodal model to calculate real-time MW_level, and based on this indicator, the robot immediately performs step-by-step interventions combining voice, gaze, and gestures in a closed-loop structure. The robot’s physical presence provides users with high salience and emotional acceptability, while the LLM dynamically generates conversational guidance tailored to the situation and user context, supporting personalized interventions. This is significant in that it goes beyond simple mindfulness support to implement a human-machine collaborative attention enhancement paradigm through real-time bio-signal-based cognitive state detection.

Based on this background, this study aims to systematically verify the effects of various real-time intervention methods on MW and attention maintenance in a mindfulness environment. To this end, we conducted a comparative experiment under the same conditions with three conditions: robot-based intervention, audio-based intervention, and no intervention. Therefore, we researched two research questions: (1) *how mindfulness environments, interventions, and MW are related*, and (2) *which interventions are effective in mindfulness*. By answering these two questions, we aim to clarify the practical benefits that a closed-loop system combining real-time biosignal analysis and LLM-driven robot feedback provides for attention enhancement and to propose design guidelines for future human-machine collaborative attention augmentation technologies.

2 Method

2.1 Experimental setup

2.1.1 Participants and Procedure. We conducted this study with 45 healthy adults (aged 19–34), recruited via public advertisements from diverse backgrounds in major, occupation, and meditation experience. Prior to the experiment, participants completed the Mindful Attention Awareness Scale (MAAS) [3] to assess baseline mindfulness, which guided balanced group assignment. Participants were evenly distributed into three groups: (G1) no intervention, (G2) real-time audio-based intervention, and (G3) real-time robot-based intervention, with 15 participants each. All participants wore EEG devices and engaged in a 20-minute mindfulness session involving breathing, body scans, and meditation. During the session, our proposed BioMW system analyzed facial expressions, posture, and eye aspect ratio (EAR) to compute MW levels in real time. When the MW level exceeded a predefined threshold, interventions were triggered for G2 and G3 only. Post-session, all participants completed a questionnaire assessing satisfaction and immersion.

2.1.2 Measurement. To analyze the relationship between various intervention measures and MW in a mindfulness environment, this study used MAAS scores, EEG analysis, and subjective satisfaction assessments as key evaluation indicators. This study used a 15-item, 6-point Likert scale version of the MAAS, with higher MAAS scores indicating higher mindfulness abilities in daily life. EEG data were recorded using the BLOSS-S8 device (BioBrain Inc.), following the international 10-20 system with eight channels (Fp1, Fp2, T7, T8, O1, O2, Fz, Pz). In particular, we focused on the Fp1 and Fp2 channels, as the prefrontal cortex was identified as the channel group most sensitive to MW and attention control fluctuations [5]. The signals were sampled at 250Hz and processed using a band-pass filter to remove low-frequency drift and high-frequency noise, and a 50Hz notch filter was applied to eliminate power line interference. Independent Component Analysis (ICA) was performed using the

MNE library to remove artifacts such as electrooculography (EOG) noise, and Fast Fourier Transform (FFT) was applied to extract alpha (8–13Hz) and beta (13–30Hz) power bands. The mind-wandering metric was defined by calculating the logarithmic sum of the Fp1-Fp2 α and β band powers and reversing the sign [10, 18]. Smaller values indicate higher $\alpha - \beta$ power, indicating higher focus and lower mind-wandering.

2.2 BioMW

The proposed BioMW model integrates real-time analysis of multi-biometric signals—facial expressions, body posture, and eye aspect ratio (EAR)—to detect MW levels and determine when intervention is needed. User images are collected frame-by-frame and pre-processed via grayscale conversion and histogram equalization to account for lighting and background variability. These are then used for facial expression recognition, posture analysis, and EAR computation. EAR is calculated based on the distance ratio between eye landmarks [12]. A persistently low EAR suggests closed eyes, potentially indicating drowsiness or reduced focus; however, fluctuation patterns are analyzed to distinguish between inattentiveness and mindful eye closure. Facial expressions are classified into seven emotional states—surprise, fear, disgust, happiness, sadness, anger, and neutral—using a ResNet-based model fine-tuned on a Korean facial expression dataset [1, 8]. Sustained boredom or excessive happiness is interpreted as increased MW. Posture is assessed via 33 body joint coordinates, with significant leaning or prolonged slouching also contributing to MW. These modality-specific signals are compared against each user’s baseline and integrated into a composite MW level:

$$MW\ level = 0.4 \times \Delta Face + 0.3 \times \Delta Pose + 0.3 \times \Delta EAR$$

Here, Δ represents the deviation from the baseline, and each modality’s contribution is weighted accordingly. Based on prior pilot tests and existing studies, the model assigns weights of $\omega_{face} = 0.4$, $\omega_{pose} = 0.3$, and $\omega_{EAR} = 0.3$. As supported by previous research, facial expressions are given the highest weight due to their strong correlation with attentional shifts during meditation. Posture, and EAR are also crucial for assessing meditation posture stability and drowsiness levels. Ultimately, the BioMW model enhances MW detection accuracy by combining facial expressions, body posture, and EAR analysis. The real-time intervention module generates appropriate feedback when the MW level exceeds a predefined threshold, as discussed in the next section.

2.3 Robot Intervention

To realise continuous human-robot interaction, the social robot Pepper was used to deliver real-time interventions during mindfulness training [17]. Equipped with voice I/O, visual sensors, and jointed motion, Pepper enabled natural interaction. Its behavior was programmed via Choregraphe and synchronized wirelessly with the experiment PC to respond instantly to BioMW outputs. When MW exceeds a threshold, the robot delivers customized interventions: weak ($0.5 \leq MW < 0.7$) or strong ($MW \geq 0.7$), ranging from gentle reminders to direct behavioral instructions. The GPT-4o-mini model generated intervention messages via two prompt types. Weak prompts emphasized attention redirection, self-awareness,

and emotional support through concise suggestions (e.g., “*Take a moment to feel your breath.*”). Strong prompts guided posture correction, specific actions, and reflection questions (e.g., “*Straighten your back and sit firmly on the ground again.*”). Ultimately, the MW level derived from the BioMW model serves as a monitoring tool and a key metric for determining the timing and intensity of interventions. The intervention content itself is dynamically generated by the GPT-4o-mini model, ensuring responsive and contextually relevant feedback.

3 Results

3.1 Results by Intervention Condition

Figure 2-(a) shows the MAAS post-test scores for the three conditions—no intervention (G1), audio intervention (G2), and robot intervention (G3)—in a box plot. The results of the one-way ANOVA indicated that the group effect was statistically significant ($p < .001$), and the post-hoc test confirmed a significant difference between G3 and G1. The mean values were approximately 58 for G1, 70 for G2, and 76 for G3, demonstrating that robot intervention maximizes subjective mindfulness improvement.

Figure 2-(b) presents the EEG-based MW_level comparison. Lower values indicate higher concentration, with $G3 < G2 < G1$, and the difference between G3 and G1 was statistically significant at the $p < .01$ level. This suggests that the robot’s multimodal feedback is most effective in suppressing attention deviation. These results empirically support RQ1, which examines the interrelationship between the mindfulness environment, intervention, and MW while also confirming the superiority of robot intervention predicted in RQ2.

3.2 MAAS-MW Relationship Analysis

Figure 2-(c) shows a scatter plot with a negative correlation coefficient between MAAS and MW_level for all 45 participants ($r = -0.78$, $p < 0.001$). This suggests that the subjective current focus experience aligns strongly with the neurophysiological indicators generated by the BioMW model. The group-specific regression lines in Figure 2-(d) show similar negative slopes across all three conditions, confirming that the sensitivity of the BioMW model remains stable regardless of the intervention condition. Notably, the intercept of the robot intervention group is the lowest, suggesting an additional phenomenon of sustained attention stabilization where MW_level is maintained at a lower level even at the same MAAS score. This pattern supports the environmental-intervention-MW linkage proposed in RQ1 while reaffirming the effectiveness of robot intervention assumed in RQ2.

3.3 Discussion

Quantitative analysis demonstrates that robot-based interventions yield the most significant results in improving mindfulness and reducing MW. This superiority is interpreted as a result of multimodal feedback and embodied social presence interacting to induce cognitive re-engagement in users. Actual participant statements provide vivid support for this mechanism.

P12 (G3) mentioned, “*When the robot looked directly at me and gently said, ‘Feel your breath at this moment,’ my scattered thoughts immediately came together.*” This shows that the combination of

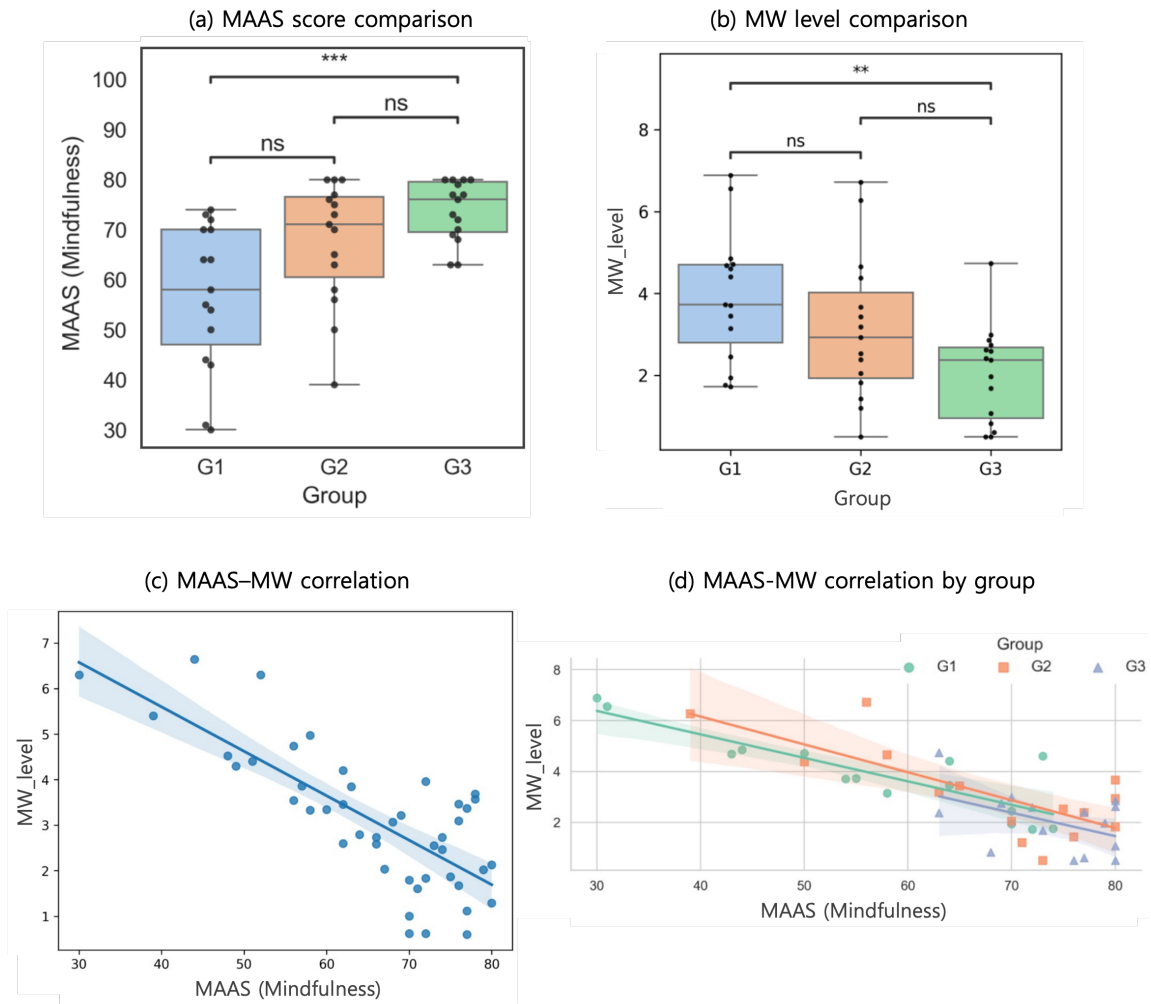


Figure 2: Statistical analyses of intervention effects.

the robot’s gaze and voice triggered an attention reset. Another participant, P41 (G3), stated, “*When the robot mimicked my posture and gestured for me to correct it, I naturally realigned my body, and at that moment, I felt my mind clear*”. This suggests that the robot’s feedback-induced sensory-based realignment quickly returned the participant from the MW state to the present moment.

In other words, this study integrated biosignal analysis, LLM-based language generation, and multimodal expression of social robots into a single closed loop to implement MW detection and customized intervention. The physical presence of the robot adds social persuasiveness to this feedback, encouraging users to monitor their attention states. This study empirically demonstrates that multimodal stimuli and embodied interaction are essential when designing attention management systems and proposes design principles that can be easily extended and applied to various environments where maintaining focus is critical, such as education, the workplace, and digital healthcare.

4 Conclusion

This study presented a real-time, closed-loop system that detects mind-wandering (MW) from facial, postural, and EAR cues and instantly delivers LLM-generated feedback through a social robot’s voice, gaze, and gestures. In a 45-participant experiment, the robot group achieved significantly higher MAAS scores ($p < .001$) and lower MW levels ($p < .01$) than both audio-only and no-intervention controls, confirming the power of multimodal, embodied prompts for attention re-engagement. These findings support the feasibility of bio-signal-driven human-machine collaboration and provide design guidance for focus-critical settings, such as classrooms, workplaces, and digital healthcare, but are limited by the study’s short duration, lab environment, small sample size, and reliance on a single prefrontal metric. Future work should extend to longer, in-the-wild trials with diverse participants, richer biosignal suites, and reinforcement-learning personalization to verify scalability and lasting impact.

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